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GitHub Website: github.com/ZitongLu1996

Zitong Lu 路子童

(Update by 09/2025)

Education & Training

Massachusetts Institute of Technology

Boston, MA, USA

Postdoctoral Research Fellow, McGovern Institute of Brain Research

2025 –

*Supported by Lore Harp McGovern 25th Anniversary Postdoctoral Fellowship

Advised by [Nancy Kanwisher](#) & [Evelina Fedorenko](#)

The Ohio State University

Columbus, OH, USA

Ph.D. in Cognitive Neuroscience, Department of Psychology

2021 – 2025

*Committee members: Julie Golomb, Andrew Leber, David Osher

Graduate Minor in Statistics, Department of Statistics

M.S. in Psychology, Department of Psychology

Advised by [Julie Golomb](#)

East China Normal University

Shanghai, China

M.S. in Cognitive Neuroscience, School of Psychology and Cognitive Science

2018 – 2021

Advised by [Yixuan Ku](#), [Yong-Di Zhou](#) & [Huimin Wang](#)

Northeastern University

Shenyang, China

B.E. in Software Engineering, Department of Software Engineering

2014 – 2018

Research Interests

(1) Neural and behavioral mechanisms of visual and semantic perception:

(A) Behavioral mechanisms of object-location binding across eye-movements; (B) Neural mechanisms of depth and 3D perception; (C) Visual and semantic representations in human brains; (D) Shared representations between visual and language modalities.

(2) Mapping representations between human brains and artificial neural networks:

(A) Brain-like artificial neural networks via neural alignment; (B) Image-to-Brain encoding models; (C) Inter-individual neural converters; (D) Reverse engineering to interpret neural mechanisms.

Using behavior, Eye-tracking, EEG, fMRI and computational methods (MVPA, RSA, IEM), and artificial deep neural networks (CNN, VAE, GAN, CLIP).

Publications

(Google Scholar: <https://scholar.google.com/citations?hl=en&user=bE5VCKsAAAAJ>)

*: (co-)first author; ♡: corresponding author

Journal articles:

23. Cheng, Y^{*ψ}, Shen, T., **Lu, Z^ψ**, Yuan, X., & Jiang, Y. (in press). The performance of deep convolutional neural networks in face recognition and comparison with the human visual system. *Journal of Psychological Science (in Chinese)*. <https://doi.org/10.16719/j.cnki.1671-6981.20250405>
22. **Lu, Z^{*ψ}**, & Wang, Y. (2025). Teaching CORnet human fMRI representations for enhanced model-brain alignment. *Cognitive Neurodynamics*, 19, 61. <https://doi.org/10.1007/s11571-025-10252-y>
21. Ran, M^{*}, **Lu, Z^ψ**, & Golomb, J.D. (2025). The influence of a moving object's location on object identity judgements. *Journal of Experimental Psychology: Human Perception and Performance*, 51(6), 764-780. <http://doi.org/10.1037/xhp0001311>
20. **Lu, Z^{*ψ}**, & Golomb, J.D. (2024 Accepted, Reviewed Preprint). Human EEG and artificial neural networks reveal disentangled representations of object real-world size in natural images. *eLife*. Reviewed preprint: <https://doi.org/10.7554/eLife.98117.1>
19. **Lu, Z^{*ψ}**, Li, W., Nie, L., & Zhao, K. (2024). A Best Practices Handbook for EEG Data Analysis with Python. *Brain-X*, 2(2), e64. <https://doi.org/10.1002/brx2.64>
18. **Lu, Z^{*ψ}**, & Golomb, J.D^ψ. (2024). Dynamic saccade context triggers more stable object-location binding. *Journal of Experimental Psychology: General*, 153(4), 873-888. (APA "Editor's Choice" paper!) <https://doi.org/10.1037/xge0001545>
17. Clayson, P.E., ..., **Lu, Z.**, ..., Langer, N. (2023 Accepted, Stage 1 Registered Replication). Contralateral delay activity as a marker of visual working memory capacity: a multi-site registered replication. *Cortex*. Preprint on *PsyArXiv*: <https://doi.org/10.31234/osf.io/shdea>
16. **Lu, Z^{*}**, & Ku, Y^ψ. (2023). Bridging the Gap between EEG and DCNNs Reveals a Fatigue Mechanism of Facial Repetition Suppression. *iScience*. 26(12), 108501. <https://doi.org/10.1016/j.isci.2023.108501>
15. **Lu, Z^{*ψ}**. (2023). Visualizing the Mind's Eye: A Future Perspective on Applications of Image Reconstruction from Brain Signals to Psychiatry. *Psychoradiology*. kkad022. <https://doi.org/10.1093/psyrad/kkad022>
14. **Lu, Z^{*}**, & Ku, Y^ψ. (2020). NeuroRA: A Python toolbox of representational analysis from multi-modal neural data. *Frontiers in Neuroinformatics*. 14: 563669. <https://doi.org/10.3389/fninf.2020.563669>

Proceedings papers:

13. Li, Y^{*ψ}, Moussa, H.N.^{*ψ}, ..., **Lu, Z.**, ..., & Sun, H^ψ. (2025). AutoSDT: Scaling Data-Driven Discovery Tasks Toward Open Co-Scientists. *Proceedings of the 2025 conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*. <https://openreview.net/forum?id=oipUPoGLPy>
12. Chen, Z^{*ψ}, Chen, S^{*ψ}, ..., **Lu, Z.**, ..., & Sun, H^ψ. (2025). ScienceAgentBench: Toward Rigorous Assessment of Language Agents for Data-Driven Scientific Discovery. *Proceedings of the 12th International Conference on Learning Representations (ICLR 2025)*. <https://openreview.net/forum?id=6z4YKr0GK6>

11. **Lu, Z^Ψ**, Wang, Y., & Golomb, J.D. (2024). ReAlnet: Achieving More Human Brain-Like Vision via Human Neural Representational Alignment. *Proceedings of the Conference on Cognitive Computational Neuroscience 2024 (CCN 2024)*.
https://2024.ccneuro.org/pdf/88_Paper_authored_ReAlnet_CCN2024_Authored.pdf
10. **Lu, Z^Ψ**, & Golomb, J.D. (2024). Probing Human Vision via an Image-to-EEG Encoding Model. *Proceedings of the Conference on Cognitive Computational Neuroscience 2024 (CCN 2024)*.
https://2024.ccneuro.org/pdf/337_Paper_authored_Img2EEG_CCN2024_Authored.pdf
9. **Lu, Z^{*}**, & Golomb, J.D. (2023). Object real-world size representations in human brains and artificial neural networks. *Proceedings of the Conference on Cognitive Computational Neuroscience 2023 (CCN 2023)*. <https://2023.ccneuro.org/proceedings/0000909.pdf>
8. **Lu, Z^{*}**, & Golomb, J.D. (2023). Generate your neural signals from mine: individual-to-individual EEG converters. *Proceedings of the 45th Annual Meeting of the Cognitive Science Society (CogSci 2023)*. <https://escholarship.org/uc/item/5xn0885t>
7. **Lu, Z^{*}**, Shafer-Skelton, A., & Golomb, J.D. (2022). Gaze-centered spatial representations in human hippocampus. *Proceedings of the Conference on Cognitive Computational Neuroscience 2022 (CCN 2022)*. <https://2022.ccneuro.org/proceedings/0000614.pdf>

Manuscripts under review / preprints:

6. **Lu, Z^Ψ**, & Golomb, J.D. (Submitted). Unfolding spatiotemporal representations of 3D visual perception in the human brain. Preprint on *bioRxiv*: <https://doi.org/10.1101/2025.08.03.668371>
5. **Lu, Z^Ψ**, Wang, Y., & Golomb, J.D. (In Revision in *Communications Biology*). Achieving More Human-Brain Like Vision via Human EEG Representational Alignment. Preprint on *Arxiv*: <https://doi.org/10.48550/arXiv.2401.17231>
4. Zeng, Y^Ψ, **Lu, Z.**, Hentz, R.E., & Osher, D.E. (Submitted). Isolated Configuration: Holistic Processing Beyond the Realm of Faces. Preprint on *bioRxiv*: <https://doi.org/10.1101/2025.03.07.641889>
3. **Lu, Z^Ψ**, & Wang, Y. (Submitted). Category-Selective Neurons in Deep Networks: Comparing Purely Visual and Visual-Language Models. Preprint on *Arxiv*: <https://doi.org/10.48550/arXiv.2502.16456>
2. Zhang, M^{*}, **Lu, Z.**, Lin, Q., Weng, X., Zhou, Y., Ma, W., Li, X., Otani, S^Ψ, & Wang, Z^Ψ. (Submitted). Transcultural differences in neural representations of the Theory of Mind between Chinese and Japanese.
1. Zhang, M^{*}, **Lu, Z^{*}**, Su, H., Kwok, S.C^Ψ, Li, X^Ψ, & Wang, Z^Ψ. (Submitted). Musical expertise attenuates cross-modal fast-“same” effect of pitches: an ERP study. Preprint on *PsyArXiv*: <https://doi.org/10.31234/osf.io/w74n>

In preparation:

- Lu, Z^Ψ**, & Golomb, J.D. (in preparation). The influence of task-irrelevant landmarks on spatiotopic localization and object-location binding.

Chiu, T-Y*[‡], **Lu, Z.**, & Golomb, J.D. (in preparation). Dynamic Read-Out of Transsaccadic Attentional Remapping from Alpha Band Oscillation.

Lu, Z*[‡], & Golomb, J.D. (in preparation). Exploring human vision through Img2EEG: An encoding framework generating high-resolution temporal EEG signals from visual inputs.

Presentations

07/15/2025 Kriegeskorte Lab, Columbia University
[Invited Talk (Virtually)] Mapping Representations between Visual Inputs, Human Brains, and Artificial Neural Networks
05/18/2025 Vision Sciences Society (VSS) 2025
[Poster] Unfolding the spatiotemporal neural mechanisms of 3D perception in the human brain: an fMRI-EEG fusion study
05/18/2025 Vision Sciences Society (VSS) 2025
[Poster] Dynamic Read-Out of Transsaccadic Attentional Remapping from Alpha Band Oscillation
12/05/2024 OSU CCBBI Research Day 2024
[Poster] Unfolding the spatiotemporal neural mechanisms of 3D perception in the human brain: an fMRI-EEG fusion study
11/22/2024 CogNeuro Prosem, OSU
[Talk] Bridging the Gap between Brain and AI in Visual Perception
10/11/2024 CogPsy Colloquium, Vrije Universiteit Amsterdam
[Invited Talk (Virtually)] Probing Human Vision: Mapping Representations between Human Brains and Artificial Neural Networks
10/02/2024 EvLab, MIT
[Invited Talk (Virtually)] Mapping Representations between Visual Inputs, Human Brains, and Artificial Neural Networks
09/13/2024 OSU CCBS Retreat 2024
[Poster] ReAlnet: Achieving More Human Brain-Like Vision via Human Neural Representational Alignment
08/09/2024 Cognitive Computational Neuroscience (CCN) 2024
[Poster] ReAlnet: Achieving More Human Brain-Like Vision via Human Neural Representational Alignment
08/07/2024 The Kanwisher Lab, MIT
[Invited Talk] Cognitive Computational Neuroscience in Visual Perception from Experiments and Computations
08/06/2024 Cognitive Computational Neuroscience (CCN) 2024
[Poster] Probing Human Vision via an Image-to-EEG Encoding Model
07/29/2024 ZJU Cognition & AI Summer School 2024
[Invited Talk (Virtually)] Exploring Human Visual Mechanisms: Integrating Human EEG and Artificial Neural Networks
06/30/2024 SJTU Brain Facts Summer School 2024
[Invited Talk (Virtually)] Bridging the Gap between Neuroscience and AI in Visual Perception
06/17/2024 School of Psychology and Cognitive Science, East China Normal University
[Invited Talk] Bridging the Gap between Neuroscience and AI in Visual Perception
06/15/2024 TCCI-NeuroChat 神聊 2024
[Invited Talk] Exploring human vision through Img2EEG: An encoding framework generating high-resolution temporal EEG signals from visual inputs
06/14/2024 School of Science, Zhejiang Sci-Tech University
[Invited Talk] Exploring Human Visual Mechanisms: Integrating Human EEG and Artificial Neural Networks
06/13/2024 School of Psychology, Nanjing Normal University
[Invited Talk] Bridging the Gap between Neuroscience and AI in Visual Perception
06/05/2024 School of Psychology, South China Normal University
[Invited Talk] Bridging the Gap between Neuroscience and AI in Visual Perception
06/04/2024 Faculty of Arts and Sciences, Beijing Normal University (Zhuhai)
[Invited Talk] Spatiotopic and retinotopic object-location binding across saccades in a more dynamic context
06/03/2024 Memory and Emotion Lab, Sun Yat-sen University
[Invited Talk] Spatiotopic and retinotopic object-location binding across saccades in a more dynamic context
05/31/2024 School of Psychology, Shenzhen University
[Invited Talk] Exploring Human Visual Mechanisms: Integrating Human EEG and Artificial Neural Networks
05/20/2024 Vision Sciences Society (VSS) 2024
[Talk] Human EEG and artificial neural networks reveal disentangled representations of object real-world size in natural images
05/19/2024 Vision Sciences Society (VSS) 2024
[Poster] Object size and depth representations in human visual cortex

03/22/2024 CogNeuro Prosem, OSU
 [Talk] Exploring human vision through Img2EEG: An encoding framework generating high-resolution temporal EEG signals from visual inputs

12/07/2023 OSU CCBBI Research Day 2023
 [Talk] Object size and depth representations in human visual cortex

12/07/2023 OSU CCBBI Research Day 2023
 [Poster] Human EEG and artificial neural networks reveal disentangled representations of object real-world size in natural images

10/18/2023 Dept of Biomedical Engineering, Tsinghua University
 [Invited Talk (virtually)] Generate your neural signals from mine: individual-to-individual EEG converters

09/22/2023 OSU CCBS Retreat 2023
 [Poster] Examining Hering's theory for color responses in human V1 and V4

09/22/2023 OSU CCBS Retreat 2023
 [Poster] The influence of a moving object's location on object identity judgments

09/22/2023 OSU CCBS Retreat 2023
 [Poster] The influence of task-irrelevant landmarks on spatiotopic localization and object-location binding

09/14/2023 Advanced Computational Neuroscience Network (ACNN) 2023
 [Poster] Generate your neural signals from mine: individual-to-individual EEG converters

08/26/2023 Cognitive Computational Neuroscience (CCN) 2023
 [Poster] Object real-world size representations in human brains and artificial neural networks

07/2023 Annual Meeting of the Cognitive Science Society (CogSci) 2023
 [Poster (virtually)] Generate your neural signals from mine: individual-to-individual EEG converters

07/2023 Eye Movements Gordon Research Conference 2023
 [Poster] Dynamic saccade context triggers more stable object-location binding

07/2023 Eye Movements Gordon Research Seminar 2023
 [Poster] Dynamic saccade context triggers more stable object-location binding

05/23/2023 Vision Sciences Society (VSS) 2023
 [Poster] A novel framework to study configural and holistic processing

05/23/2023 Vision Sciences Society (VSS) 2023
 [Poster] Examining Hering's theory for color responses in human V1 and V4

05/21/2023 Vision Sciences Society (VSS) 2023
 [Poster] The influence of a moving object's location on object identity judgments

05/20/2023 Vision Sciences Society (VSS) 2023
 [Poster] The influence of task-irrelevant landmarks on spatiotopic localization and object-location binding

04/21/2023 OSU CCBBI Student Workshop
 [Talk] Decode brain representations based on Python

03/31/2023 CogNeuro Prosem, OSU
 [Talk] Generate your neural signals from mine: individual-to-individual EEG converters

03/08/2023 CogPsy Prosem, OSU
 [Talk] Object real-world size representations in human brains and artificial neural networks

12/31/2022 The 2nd Neural Network Interdisciplinary Forum 2022
 [Talk (virtually)] Facial representation comparisons between human brain and hierarchical deep convolutional neural network reveal a fatigue repetition suppression mechanism

11/12/2022 Society of Neuroscience (SFN) 2022
 [Poster] Dynamic saccade context triggers more stable object-location binding

10/22/2022 OSU CCBS Retreat 2022
 [Talk] Dynamic saccade context triggers more stable object-location binding

08/27/2022 Cognitive Computational Neuroscience (CCN) 2022
 [Poster] Gaze-centered spatial representations in human hippocampus

05/17/2022 Vision Sciences Society (VSS) 2022
 [Talk] Dynamic saccade context triggers spatiotopic object-location binding

04/15/2022 CogNeuro Prosem, OSU
 [Talk] Dynamic saccade context triggers spatiotopic object-location binding

12/06/2021 OSU CCBBI Research Day 2021
 [Talk] Gaze-centered spatial representations in human hippocampus

11/11/2021 Society of Neuroscience (SFN) 2021
 [Poster (virtually)] Representation comparisons between human brain and hierarchical deep convolutional neural network in face perception reveal a fatigue mechanism of repetition suppression

08/27/2021	The European Conference on Visual Perception (ECVP) 2021
[Poster (virtually)] Representation comparisons between human brain and hierarchical deep convolutional neural network in face perception reveal a fatigue mechanism of repetition suppression	
06/27/2021	Centre for Cognition and Brain Science, University of Macau
[Invited Talk (virtually)] Using computational methods to explore the neural representational mechanism in cognitive neuroscience	
06/08/2021	UNIQUE Student Symposium 2021
[Talk (virtually)] Representation comparisons between human brain and hierarchical deep convolutional neural network in face perception reveal a fatigue mechanism of repetition suppression	
12/28/2020	Dept of Biomedical Engineering, UESTC
[Invited Talk (virtually)] Representational Analysis for Cognitive Neuroscience based on NeuroRA	
12/17/2020	Dept of Biomedical Engineering, Shenzhen University
[Invited Talk] Representational Analysis for Cognitive Neuroscience based on NeuroRA	

Toolboxes & Tutorials

NeuroRA toolbox:

<https://zitonglu1996.github.io/NeuroRA> (186 stars on GitHub!)

Citation: Lu, Z., & Ku, Y. (2020). NeuroRA: A Python toolbox of representational analysis from multi-modal neural data. *Frontiers in Neuroinformatics*. 14:563669. <https://doi.org/10.3389/fninf.2020.563669>

Python EEG data analysis handbook:

<https://github.com/ZitongLu1996/Python-EEG-Handbook> (69 stars on GitHub!)

Chinese version:

<https://github.com/ZitongLu1996/Python-EEG-Handbook-CN> (464 stars on GitHub!)

EEG2EEG:

<https://github.com/ZitongLu1996/EEG2EEG> (28 stars on GitHub!)

Citation: Lu, Z., & Golomb, J.D. (2023). Generate your neural signals from mine: individual-to-individual EEG converters. *Proceedings of the 45th Annual Meeting of the Cognitive Science Society (CogSci 2023)*.
<https://escholarship.org/uc/item/5xn0885t>

Programming & Experiment Skills

Computer Languages: Python, MATLAB, C, C++, Java, Julia

Software & Toolboxes: Psychtoolbox, EEGLAB, MNE-Python, BrainVogager, Freesurfer, Nibabel, Nilearn, NeuroRA, PyTorch

Experimental Experience: EEG, fMRI, and Eye tracker

Working Experience

08/2022-08/2025

OSU EEG lab manager

Dept of Psychology, OSU

01/2024-08/2025

EEG Tech Search Committee

Dept of Psychology, OSU

08/2023-08/2025

CCBBI Technical Committee

Center for Cognitive and Behavioral Brain Imaging, OSU

08/2023-12/2023

Teaching Assistant (Co-Instructor) (PSYCH 5621 - Intro to ERP)

Dept of Psychology, OSU

08/2023-12/2023

Teaching Assistant (PSYCH 3310 – Sensation and Perception)

Dept of Psychology, OSU

08/2022-04/2023

Teaching Assistant (PSYCH 4510 - Cognitive Psychology Lab)

Dept of Psychology, OSU

09/2020-06/2021

Research Assistant

Brain-Like Memory Group, Peng Cheng Laboratory

05/2017-08/2017

Programmer (as Project Leader)

iSoftStone corporation

Mentoring

Mengxin Ran (undergrad student at The Ohio State University; now lab manager at Brown University w/ Oriel Feldmanhall)

Wanru Li (undergrad student at East China Normal University; now PhD student at Peking University w/ Pinglei Bao)

Honors & Awards

- 2025-2027 Lore Harp McGovern 25th Anniversary Postdoctoral Fellowship (USD 100,000 * 2 years, MIT)
- 2025 Summer Research Excellence Awards (USD 6,500, OSU Dept of Psychology)
- 2024 Summer Research Excellence Awards (USD 6,500, OSU Dept of Psychology)
- 2022 CCBBI Gibson Research Award (USD 3,000, OSU CCBBI)
- 2021-2022 University Fellowship (USD 30,000, OSU)
- 2021 Outstanding Graduate Student (3%, Department of Education of Shanghai City)
- 2019-2020 Short-Term Overseas Research Scholarship (about USD 7,000, by ECNU)
- 2018 Third prize (30%, China Graduate Student Mathematical Contest in Modeling)
- 2017 Outstanding Graduate Student (3%, Department of Education of Liaoning Province)
- 2017 Second-Class Merit Scholarship (13%, by NEU)
- 2017 Meritorious Winner (13%, Mathematical Contest in Modeling, by the U.S COMAP)
- 2016 First-Class Liu Dajie & Fang Wenyu's Scholarship (<1%, USD 1500, by NEU)
- 2016 Provincial First Prize (3%, China Undergraduate Mathematical Contest in Modeling)
- 2016 First-Class Merit Scholarship (3%, by NEU)
- 2016 Honorable Mention (30%, Mathematical Contest in Modeling, by the U.S COMAP)
- 2015 Second-Class Merit Scholarship (13%, by NEU)

Ad Hoc Reviewer

Scientific Reports; Advances in Psychological Science; Conference on Cognitive Computational Neuroscience (CCN) 2022; Annual meeting of the cognitive science society (CogSci) 2023; Conference on Cognitive Computational Neuroscience (CCN) 2023; Cerebral Cortex; Experimental Brain Research; Heliyon; Neural Networks; ICLR Re-Algn Workshop (2025); Conference on Cognitive Computational Neuroscience (CCN) 2025; Communications Biology; Frontiers in Neuroinformatics; Attention, Perception, & Psychophysics; Cognitive Processing; Plos Computational Biology.

Collaborators & Close Academic Friends

Yuxin Wang – University of Cincinnati

Yile Wang – The University of Texas at Dallas

Wanru Li – Peking University
Dr. Fan Cheng – Columbia University
Tzu-Yao Chiu – The Ohio State University
Yong Min Choi – The Ohio State University
Tianyu Zhang – The Ohio State University
Dr. David Osher – The Ohio State University
Yuxuan Zeng – The Ohio State University
Dr. Jin Li – Georgia Institute of Technology
Dr. Jiaqi Li – Peking University & University of Oxford
Xianhui He – University of Oxford
Mengxin (AvA) Ran – Brown University
Eva Lout – University of Texas at Austin
Mingmin Zhang – University of Groningen
Dr. Yixuan Ku – Sun Yat-sen University
Lu Nie – Sun Yat-sen University
Dr. Qingrong Chen – Nanjing Normal University
Dr. Jingjing Zhang – Nanjing Normal University
Dr. Dongwei Li – Beijing Normal University